Executive Summary

This analysis explores customer characteristics influencing treadmill purchases at AeroFit. The study identifies three distinct customer segments based on income, fitness level, and usage behavior. Results show that behavioral factors are stronger predictors of product choice than demographic attributes. The findings support targeted marketing, personalized recommendations, and upgrade strategies to improve sales performance.

Problem Statement

Aerofit aims to understand the characteristics of customers purchasing different treadmill models (KP281, KP481, KP781).

The objective of this analysis is to:

- Explore customer demographic and fitness attributes
- Identify differences among buyers of each treadmill product
- Build customer profiles
- Compute probabilities to support better product recommendations for new customers.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

df = pd.read_csv("aerofit_treadmill.csv")
df.head()
```

{"summary":"{\n \"name\": \"df\",\n \"rows\": 180,\n \"fields\": [\n {\n \"column\": \"Product\",\n \"properties\": {\n \"dtype\": \"category\",\n \"num_unique_values\": 3,\n \"samples\": [\n \"KP281\",\n \"KP481\",\n \"KP781\"\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n },\n {\n \"column\": \"Age\",\n \"properties\": {\n \"dtype\": \"number\",\n \"std\": 6,\n \"min\": 18,\n \"max\": 50,\n \"num_unique_values\": 32,\n \"samples\": [\n 45,\n 33,\n 43\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n },\n {\n \"column\": \"Gender\",\n \"properties\": {\n \"dtype\": \"category\",\n \"num_unique_values\": 2,\n \"samples\": [\n \"Female\",\n \"Male\"\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n },\n {\n \"column\": \"Education\",\n \"properties\": {\n \"dtype\": \"number\",\n \"std\": 1,\n \"min\": 12,\n \"max\": 21,\n \"num_unique_values\": 8,\n \"samples\": [\n 15,\n 18\n],\n \"semantic_type\": \"\",\n

```
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"MaritalStatus\",\n      \"properties\": {\n        \"dtype\":
\"category\",\n        \"num_unique_values\": 2,\n        \"samples\":
[\n          \"Partnered\",\n          \"Single\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Usage\",\n      \"properties\": {\
n        \"dtype\": \"number\",\n        \"std\": 1,\n        \"min\":
2,\n        \"max\": 7,\n        \"num_unique_values\": 6,\n
\"samples\": [\n          3,\n          2\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Fitness\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\": 0,\n
\"min\": 1,\n        \"max\": 5,\n        \"num_unique_values\": 5,\n
\"samples\": [\n          3,\n          5\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Income\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\": 16506,\n
\"min\": 29562,\n        \"max\": 104581,\n
\"num_unique_values\": 62,\n        \"samples\": [\n          88396,\n
103336\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"Miles\",\n      \"properties\": {\n        \"dtype\": \"number\",\n
\"std\": 51,\n        \"min\": 21,\n        \"max\": 360,\n
\"num_unique_values\": 37,\n        \"samples\": [\n          95,\n
169\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"df"}
```


The dataset contains customer purchase information including demographic details, income, fitness level, and treadmill product purchased.

```
df.shape
```

```
(180, 9)
```

The dataset contains 180 rows and 9 columns representing customer characteristics and purchase behavior.

```
df.columns
```

```
Index(['Product', 'Age', 'Gender', 'Education', 'MaritalStatus',
'Usage',
       'Fitness', 'Income', 'Miles'],
      dtype='object')
```

The dataset includes product information along with demographic, financial, and fitness-related variables.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Product        180 non-null    object
 1   Age            180 non-null    int64
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64
 6   Fitness        180 non-null    int64
 7   Income         180 non-null    int64
 8   Miles          180 non-null    int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

Observations:

- Product, Gender and MaritalStatus are categorical variables.
- Age, Education, Usage, Income, Fitness, and Miles are numerical variables.
- No missing values are observed in the dataset.

```
categorical_cols = ["Product", "Gender", "MaritalStatus"]

for col in categorical_cols:
    df[col] = df[col].astype("category")

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Product        180 non-null    category
 1   Age            180 non-null    int64
 2   Gender         180 non-null    category
 3   Education      180 non-null    int64
 4   MaritalStatus  180 non-null    category
 5   Usage          180 non-null    int64
 6   Fitness        180 non-null    int64
 7   Income         180 non-null    int64
 8   Miles          180 non-null    int64
dtypes: category(3), int64(6)
memory usage: 9.5 KB
```

Categorical variables were converted to 'category' datatype to improve memory efficiency and enable better categorical analysis.

```
df.describe()
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 8,\n  \"fields\": [\n  {\n  \"column\": \"Age\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 55.58832332198464,\n  \"min\": 6.943498135399795,\n  \"max\": 180.0,\n  \"num_unique_values\": 8,\n  \"samples\": [\n  28.788888888888888,\n  26.0,\n  180.0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Education\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 59.04362112875324,\n  \"min\": 1.6170548978065553,\n  \"max\": 180.0,\n  \"num_unique_values\": 7,\n  \"samples\": [\n  180.0,\n  15.572222222222223,\n  16.0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Usage\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 62.474604277313155,\n  \"min\": 1.0847970343962445,\n  \"max\": 180.0,\n  \"num_unique_values\": 7,\n  \"samples\": [\n  180.0,\n  3.4555555555555557,\n  4.0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Fitness\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 62.63086276036247,\n  \"min\": 0.958868565619312,\n  \"max\": 180.0,\n  \"num_unique_values\": 7,\n  \"samples\": [\n  180.0,\n  3.311111111111111,\n  4.0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Income\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 31403.855763201762,\n  \"min\": 180.0,\n  \"max\": 104581.0,\n  \"num_unique_values\": 8,\n  \"samples\": [\n  53719.57777777778,\n  50596.5,\n  180.0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Miles\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 106.52090041797726,\n  \"min\": 21.0,\n  \"max\": 360.0,\n  \"num_unique_values\": 8,\n  \"samples\": [\n  103.19444444444444,\n  94.0,\n  180.0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  }\n  ]\n}","type":"dataframe"}
```


Key Observations from Statistical Summary:

Age: Most of our customers are quite young, around 29 years old on average.

Usage: People generally plan to use their treadmills about 3 to 4 times a week.

Income: The typical customer earns around $53,000, which means we have a lot of middle-income buyers.

Fitness: On a scale of 1 to 5, the average fitness level is about 3, suggesting our users are moderately fit.

Miles: The number of miles people run weekly varies a lot, from those who walk casually to serious runners.

```
df.describe(include ='all')
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 11,\n  \"fields\": [\n
{\n      \"column\": \"Product\",\n      \"properties\": {\n
\"dtype\": \"category\",\n        \"num_unique_values\": 4,\n
\"samples\": [\n          3,\n          \"80\",\n          \"180\"\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    },\n    {\n      \"column\": \"Age\",\n      \"properties\": {\
n        \"dtype\": \"number\",\n        \"std\": 55.58832332198464,\n
\"min\": 6.943498135399795,\n        \"max\": 180.0,\n
\"num_unique_values\": 8,\n        \"samples\": [\n
28.788888888888888,\n          26.0,\n          180.0\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Gender\",\n      \"properties\":
{\n        \"dtype\": \"category\",\n        \"num_unique_values\":
4,\n        \"samples\": [\n          2,\n          \"104\",\n
\"180\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"Education\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 59.04362112875324,\n        \"min\":
1.6170548978065553,\n        \"max\": 180.0,\n
\"num_unique_values\": 7,\n        \"samples\": [\n          180.0,\n
15.572222222222223,\n          16.0\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"MaritalStatus\",\n
\"properties\": {\n        \"dtype\": \"category\",\n
\"num_unique_values\": 4,\n        \"samples\": [\n          2,\n
\"107\",\n          \"180\"\n        ],\n        \"semantic_type\":
\"\",\n        \"description\": \"\"\n      }\n    },\n    {\n
\"column\": \"Usage\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 62.474604277313155,\n        \"min\":
1.0847970343962445,\n        \"max\": 180.0,\n
\"num_unique_values\": 7,\n        \"samples\": [\n          180.0,\n
3.4555555555555557,\n          4.0\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Fitness\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
62.63086276036247,\n        \"min\": 0.958868565619312,\n
\"max\": 180.0,\n        \"num_unique_values\": 7,\n
\"samples\": [\n          180.0,\n          3.311111111111111,\n
4.0\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"Income\",\n      \"properties\": {\n        \"dtype\": \"number\",\n
\"std\": 31403.855763201762,\n        \"min\": 180.0,\n
```

\"max\": 104581.0,\n   \"num_unique_values\": 8,\n
\"samples\": [\n   53719.57777777778,\n   50596.5,\n
180.0\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
\"Miles\",\n   \"properties\": {\n   \"dtype\": \"number\",\n
\"std\": 106.52090041797726,\n   \"min\": 21.0,\n   \"max\":
360.0,\n   \"num_unique_values\": 8,\n   \"samples\": [\n
103.19444444444444,\n   94.0,\n   180.0\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   }\n]\n}","type":"dataframe"}

The KP281 treadmill has the highest purchase frequency among all products, indicating strong demand for entry-level equipment.

2.Non-Graphical Analysis: Value Counts & Unique Attributes (10 Marks)

```
df.nunique()
```

```
Product           3
Age              32
Gender            2
Education         8
MaritalStatus     2
Usage             6
Fitness           5
Income           62
Miles            37
dtype: int64
```

Observations:

- The dataset contains 3 treadmill products.
- Gender has 2 categories: Male and Female.
- MaritalStatus has 2 categories: Single and Partnered.
- Numerical variables show sufficient variation, indicating diverse customer characteristics.

Unique values per column

```
for col in df.columns:
    print(col, ":", df[col].unique())
```

```
Product : ['KP281', 'KP481', 'KP781']
Categories (3, object): ['KP281', 'KP481', 'KP781']
Age : [18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38
39 40 41
 43 44 46 47 50 45 48 42]
Gender : ['Male', 'Female']
Categories (2, object): ['Female', 'Male']
```

```
Education : [14 15 12 13 16 18 20 21]
MaritalStatus : ['Single', 'Partnered']
Categories (2, object): ['Partnered', 'Single']
Usage : [3 2 4 5 6 7]
Fitness : [4 3 2 1 5]
Income : [ 29562  31836  30699  32973  35247  37521  36384  38658
40932  34110
  39795  42069  44343  45480  46617  48891  53439  43206  52302  51165
  50028  54576  68220  55713  60261  67083  56850  59124  61398  57987
  64809  47754  65220  62535  48658  54781  48556  58516  53536  61006
  57271  52291  49801  62251  64741  70966  75946  74701  69721  83416
  88396  90886  92131  77191  52290  85906 103336  99601  89641  95866
 104581  95508]
Miles : [112  75  66  85  47 141 103  94 113  38 188  56 132 169  64
53 106  95
 212  42 127  74 170  21 120 200 140 100  80 160 180 240 150 300 280
260
 360]
```

Each categorical variable contains clearly defined categories with no inconsistent labels, indicating clean data.

Product Purchase distribution

```
df["Product"].value_counts()

df["Product"].value_counts(normalize=True)

Product
KP281    0.444444
KP481    0.333333
KP781    0.222222
Name: proportion, dtype: float64
```

Observations:

- KP281 has the highest number of purchases.
- KP481 has moderate sales.
- KP781 has the lowest purchase frequency.

This indicates that customers prefer entry-level treadmills over premium models, suggesting price sensitivity in the market.

Gender Distribution

```
df["Gender"].value_counts()
df["Gender"].value_counts(normalize=True)

Gender
Male      0.577778
```

```
Female    0.422222
Name: proportion, dtype: float64
```

Male customers form the majority of treadmill buyers, accounting for more than half of total purchases.

```
df["MaritalStatus"].value_counts()
```

```
MaritalStatus
Partnered    107
Single        73
Name: count, dtype: int64
```

Partnered customers represent a larger share of buyers compared to single customers, suggesting treadmill purchases are common among households and families.

```
df["Education"].value_counts().sort_index()
```

```
Education
12     3
13     5
14    55
15     5
16    85
18    23
20     1
21     3
Name: count, dtype: int64
```

Customers have varying education levels, indicating that treadmill purchases are not limited to a specific education group.

```
df["Fitness"].value_counts().sort_index()
```

```
Fitness
1     2
2    26
3    97
4    24
5    31
Name: count, dtype: int64
```

Most customers rate their fitness between 2 and 4, showing that AeroFit products attract moderately active individuals rather than only elite athletes.

Usage frequency distribution

```
df["Usage"].value_counts().sort_index()
```

```
Usage
2    33
3    69
4    52
5    17
6     7
7     2
Name: count, dtype: int64
```

Most customers plan to use the treadmill 3–4 times per week, indicating regular but not extreme usage behavior.

Key Insights:

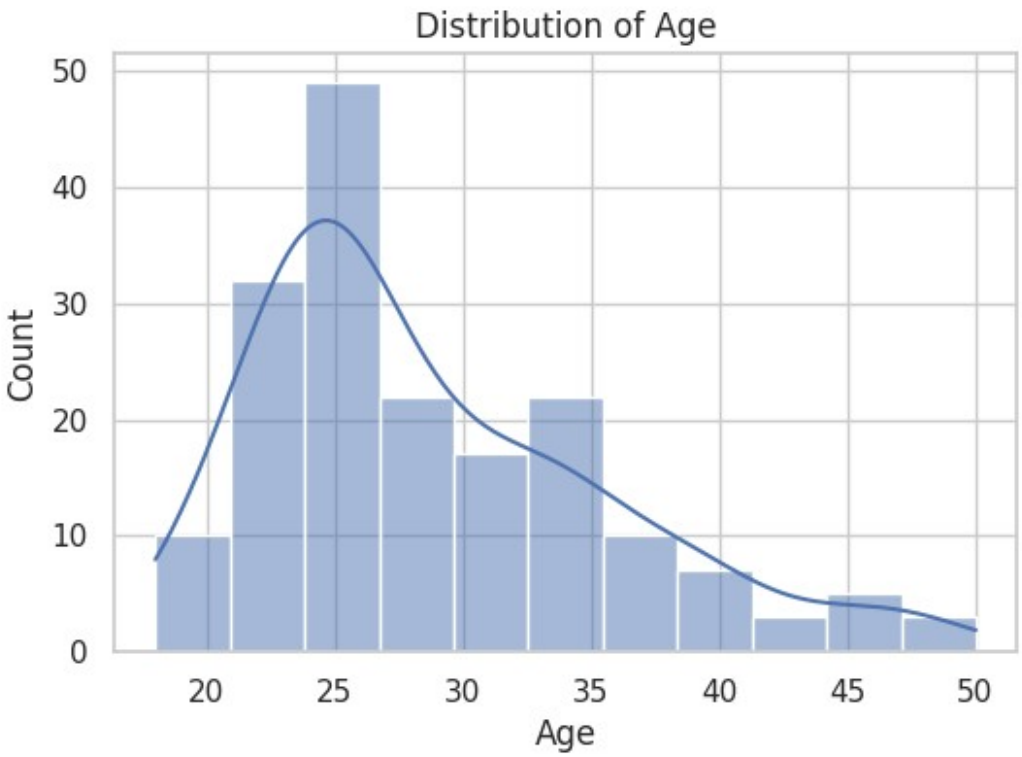
- Entry-level treadmill KP281 drives the majority of sales.
- The customer base is slightly male-dominated.
- Partnered individuals form a major customer segment.
- Most buyers consider themselves moderately fit.
- Typical users plan consistent weekly exercise rather than intensive training.

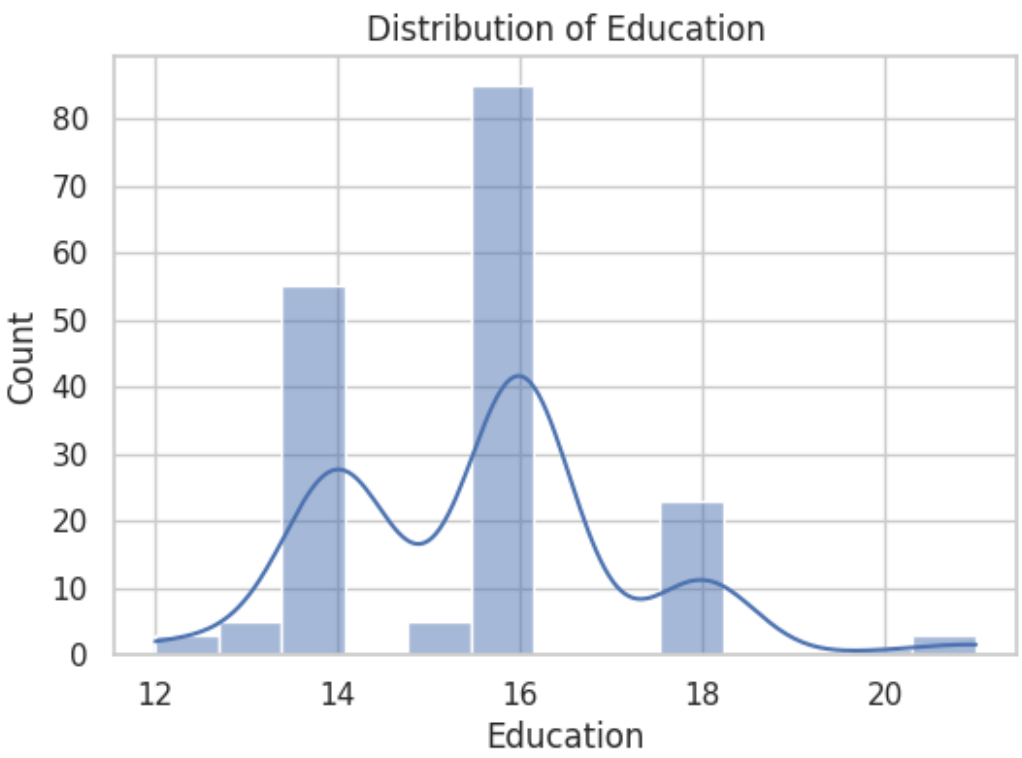
3.Visual Analysis - Univariate & Bivariate:

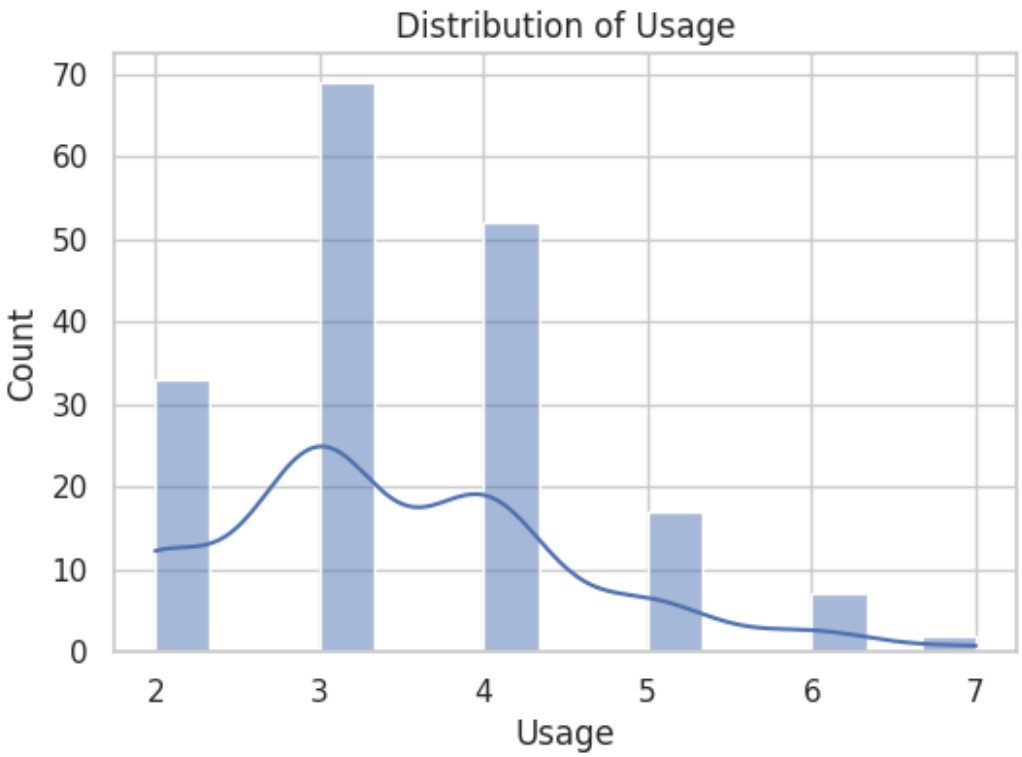
A.Univariate Analysis(Continous variable)

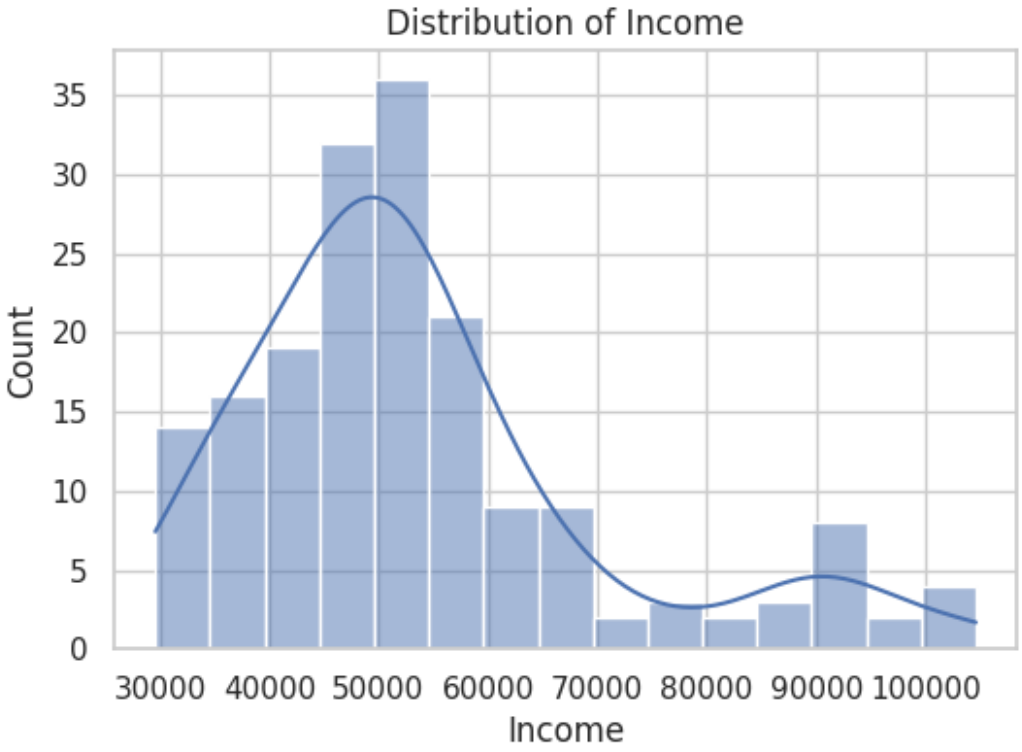
1.Histograms + Distribution Plots Continous variables Age Education Usage Income Fitness Miles

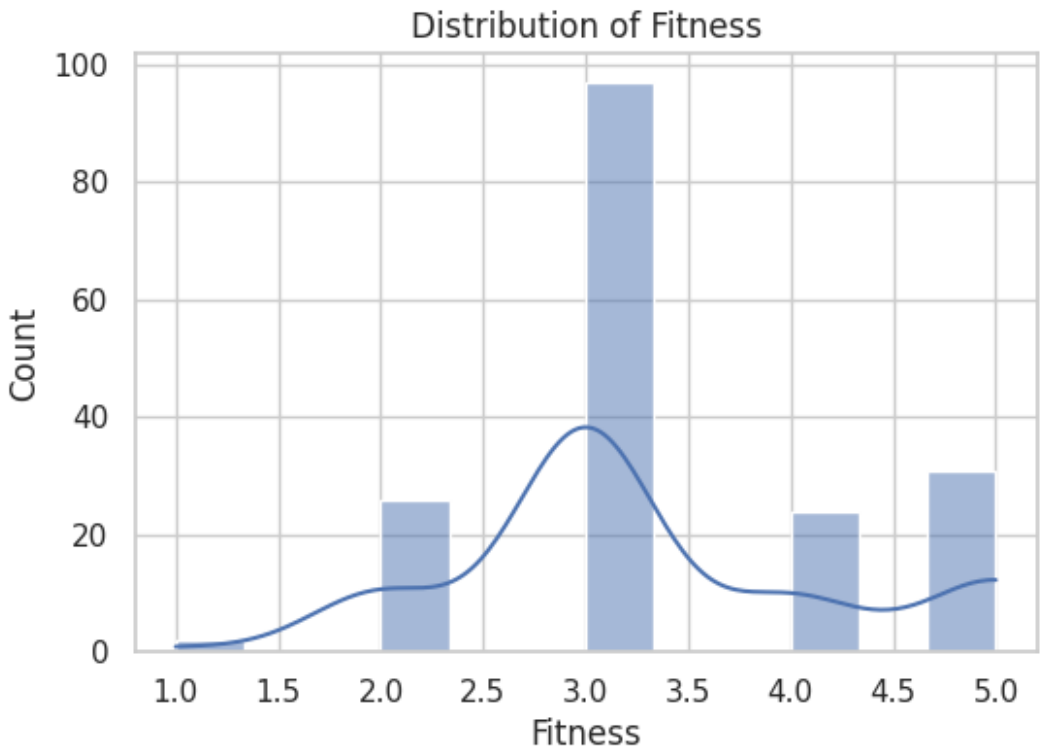
```
num_cols = ["Age","Education","Usage","Income","Fitness","Miles"]

for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.show()
```

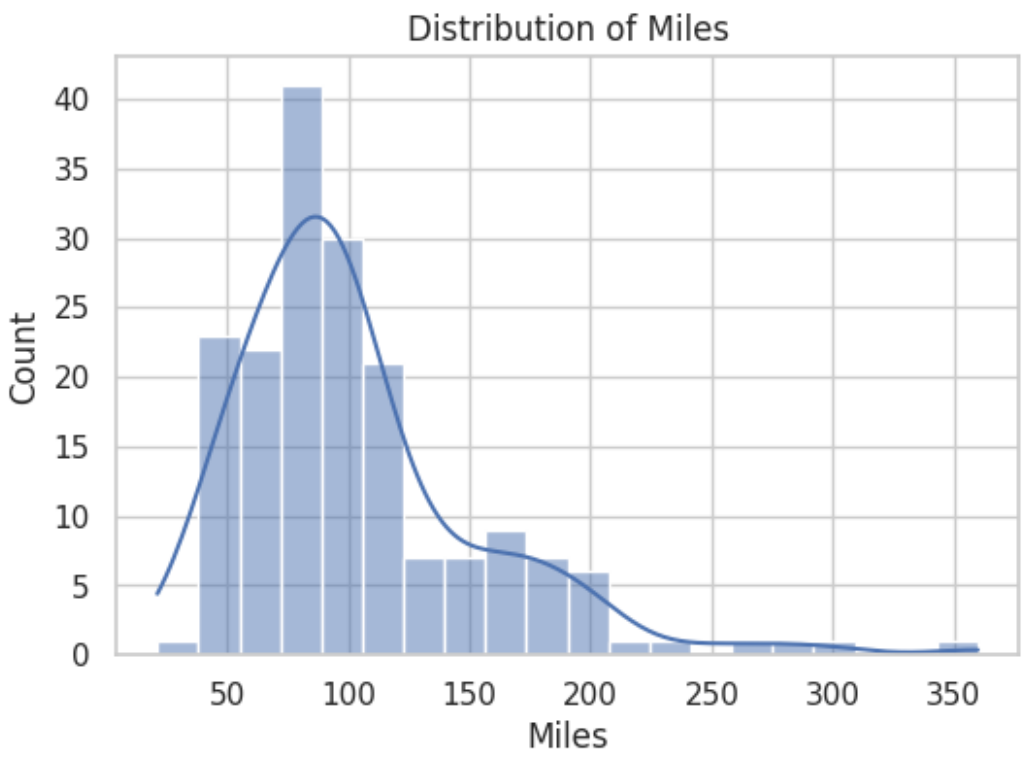

Distribution of Age
Count
Age
0
10
20
30
40
50
20
25
30
35
40
45
50


Distribution of Education
Count
Education
0
10
20
30
40
50
60
70
80
12
14
16
18
20


Distribution of Usage
Count
Usage
0
10
20
30
40
50
60
70
2
3
4
5
6
7


Distribution of Income
Count
Income
0
5
10
15
20
25
30
35
30000
40000
50000
60000
70000
80000
90000
100000





Age: Age distribution shows most customers are between 20–35 years, indicating a young customer base interested in home fitness equipment.

Education: Education levels are concentrated around mid-range values, suggesting treadmill buyers come from diverse educational backgrounds.

Usage: Most customers plan to use treadmills 3–4 times per week, indicating consistent but moderate exercise habits.

Income: Income distribution is right-skewed, with a small group of high-income customers contributing to premium product purchases.

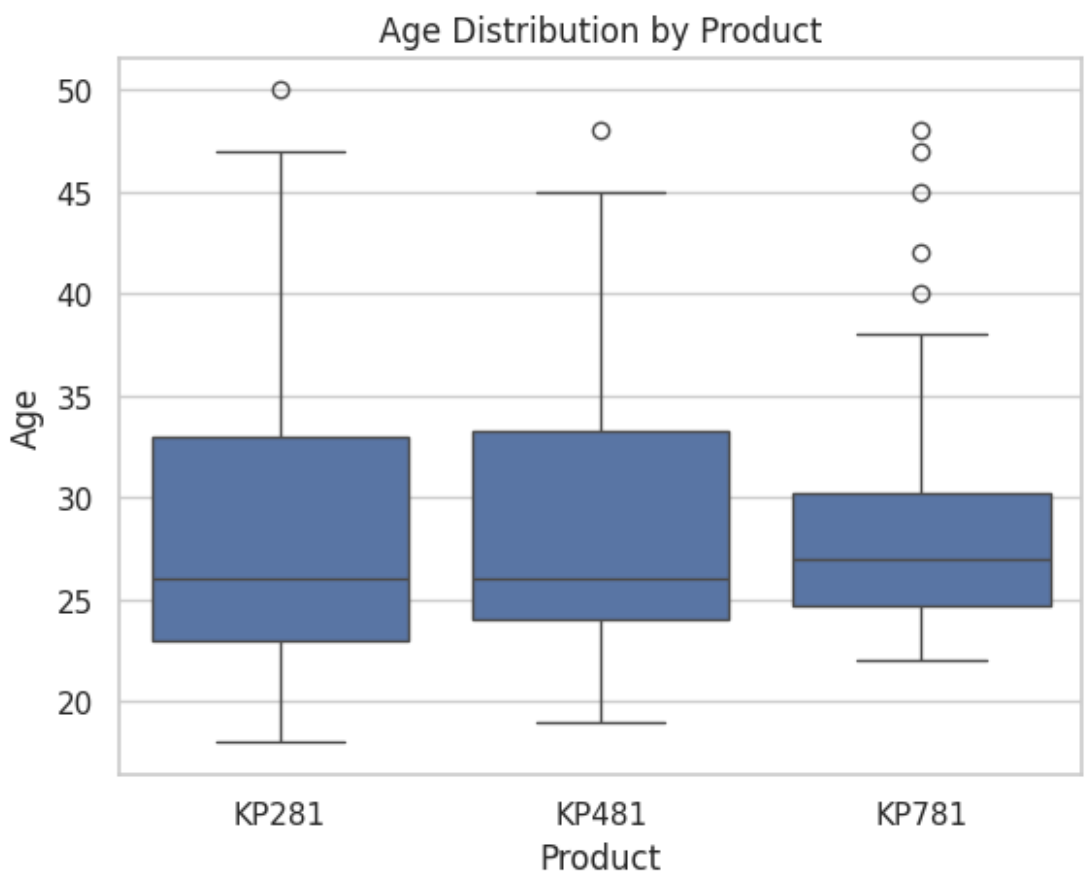
Fitness: Fitness ratings are centered around level 3, showing most buyers consider themselves moderately fit.

Miles: Miles walked/run per week shows wide variation, indicating both casual users and serious runners exist in the dataset.

B. Bivariate Analysis using Boxplots

1.Product vs Age

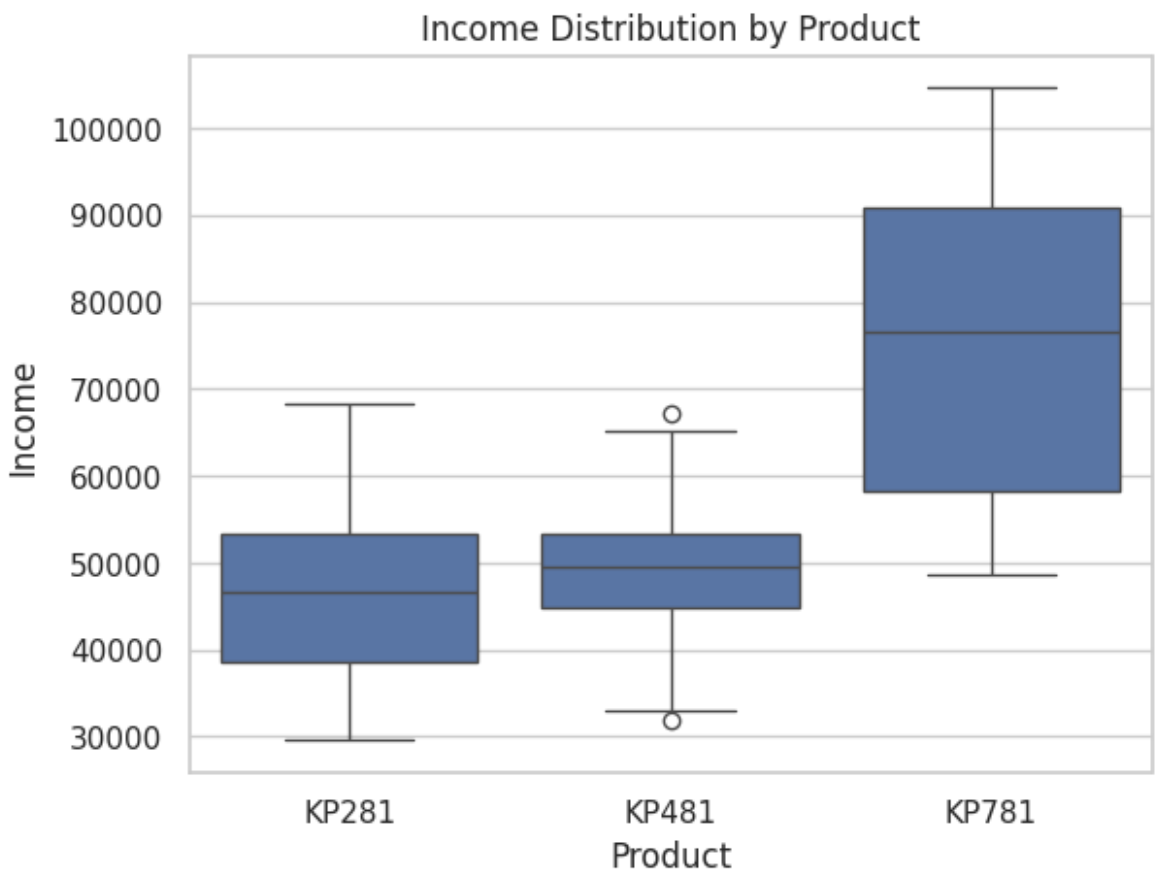
```
import seaborn as sns
sns.boxplot(data=df, x="Product", y="Age")
plt.title("Age Distribution by Product")
plt.show()
```



Age distribution is similar across products, suggesting age is not a strong determinant of treadmill choice.

2.Product vs Income

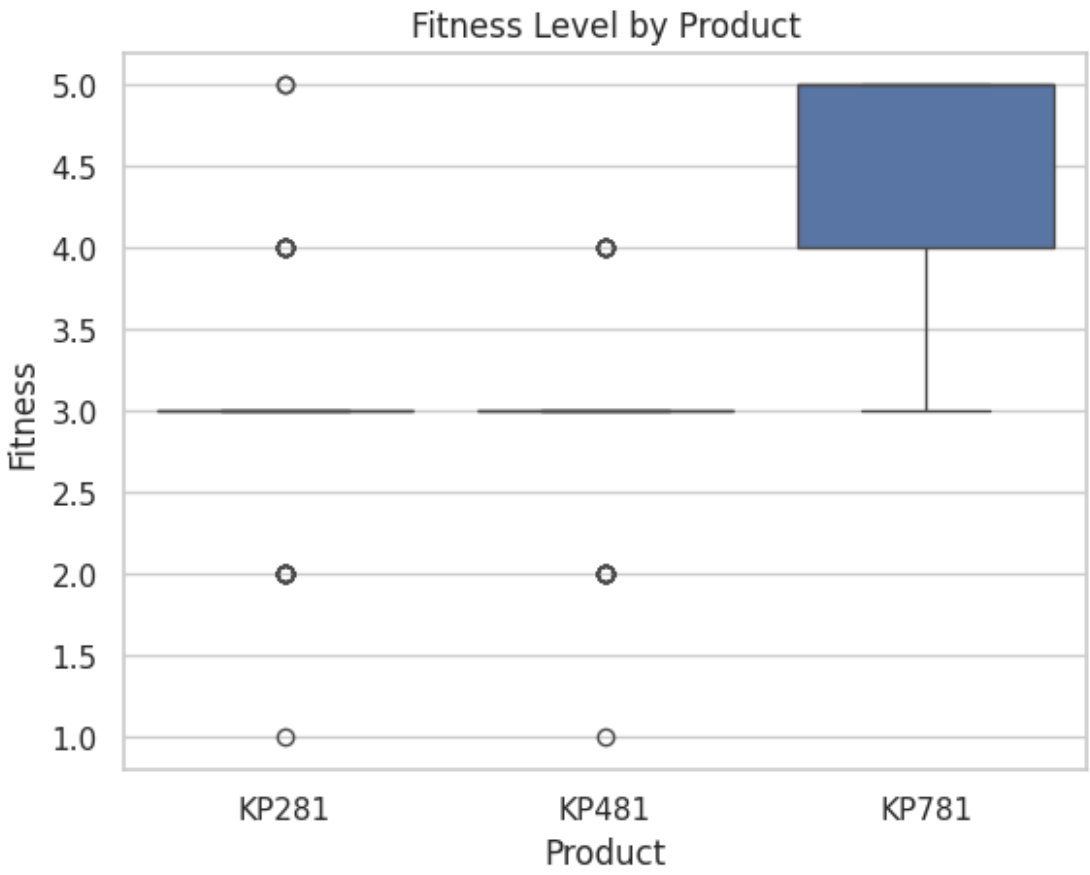
```
sns.boxplot(data=df, x="Product", y="Income")
plt.title("Income Distribution by Product")
plt.show()
```



Customers purchasing KP781 have significantly higher income levels compared to KP281 and KP481 buyers, indicating premium products attract wealthier customers.

3.Product vs Fitness

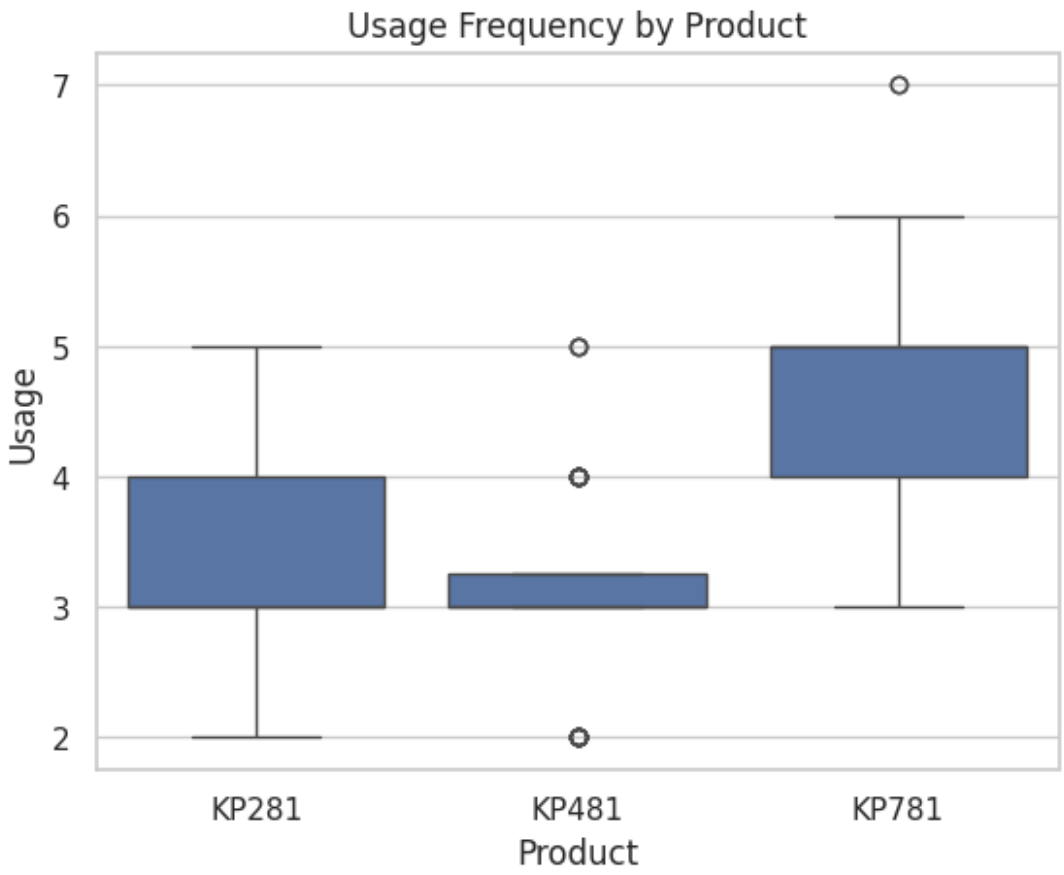
```
sns.boxplot(data=df, x="Product", y="Fitness")
plt.title("Fitness Level by Product")
plt.show()
```



KP781 buyers show higher fitness ratings, suggesting advanced treadmills are preferred by highly active individuals.

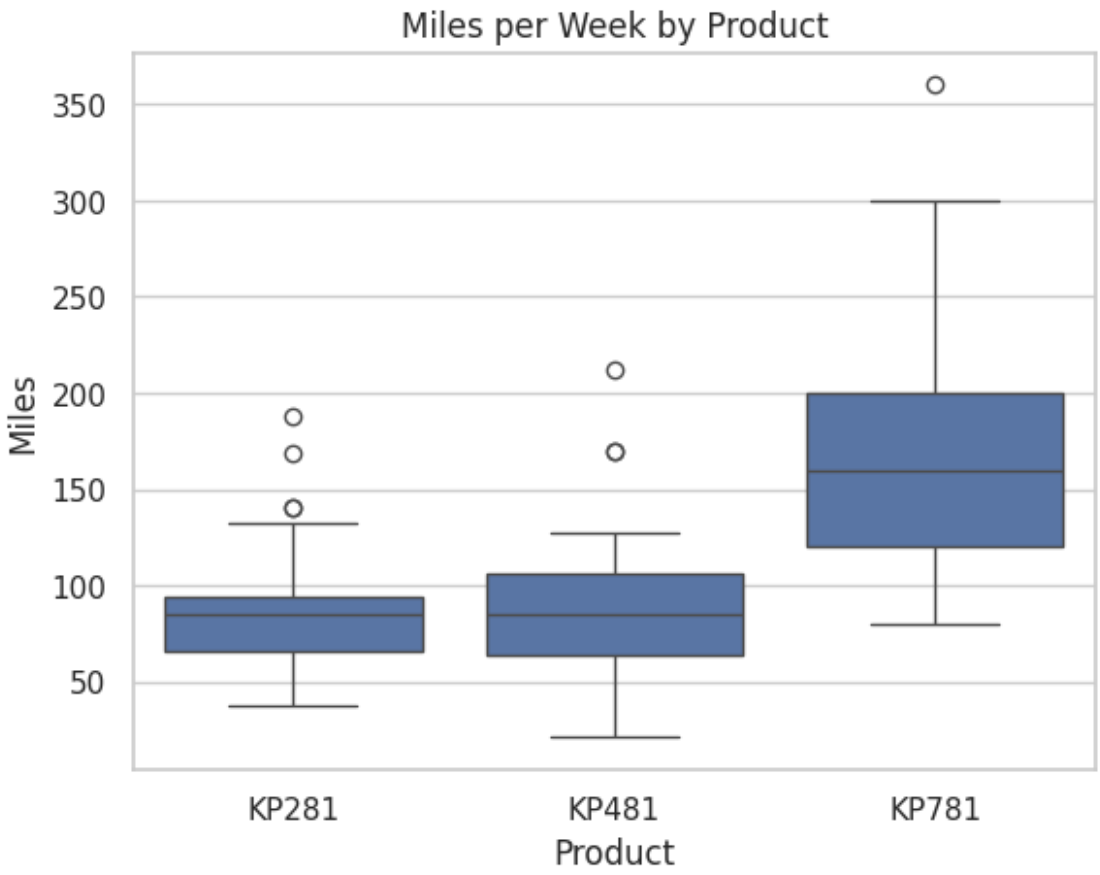
4.Product vs Usage

```
sns.boxplot(data=df, x="Product", y="Usage")
plt.title("Usage Frequency by Product")
plt.show()
```



5.Product Vs Miles

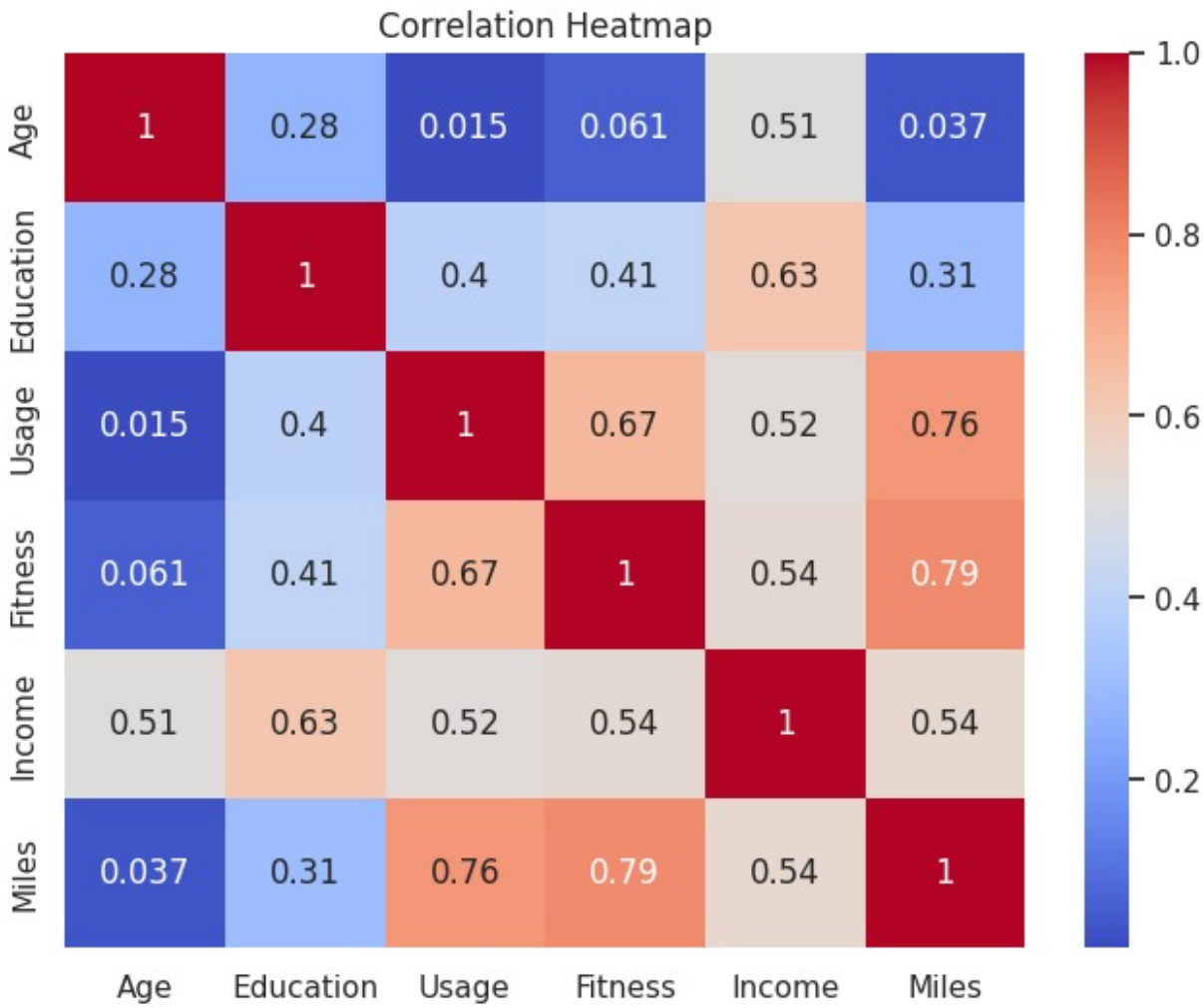
```
sns.boxplot(data=df, x="Product", y="Miles")
plt.title("Miles per Week by Product")
plt.show()
```



KP781 buyers expect to run significantly more miles weekly, reinforcing that premium products target serious runners.

C.Correlation Analysis

```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



Fitness and Miles show strong positive correlation, indicating fitter individuals run more weekly miles. Usage also positively correlates with Miles, suggesting frequent users engage in longer workouts. Income shows moderate association with premium purchase behavior indirectly.

Key Visual Insights:

- Income, fitness level, and miles are strong differentiators among treadmill buyers.
- Premium treadmill KP781 is associated with higher income and advanced fitness users.
- Age does not significantly influence product choice.
- Customer behavior variables (usage and miles) are better predictors than demographics.

4.Missing value

```
df.isnull().sum()

Product          0
Age              0
Gender           0
Education        0
```

```
MaritalStatus    0
Usage            0
Fitness          0
Income           0
Miles            0
dtype: int64
```

Observations:

- No missing values are present in the dataset.
- The dataset is complete and suitable for analysis without data removal.

2.Percentage of missing values

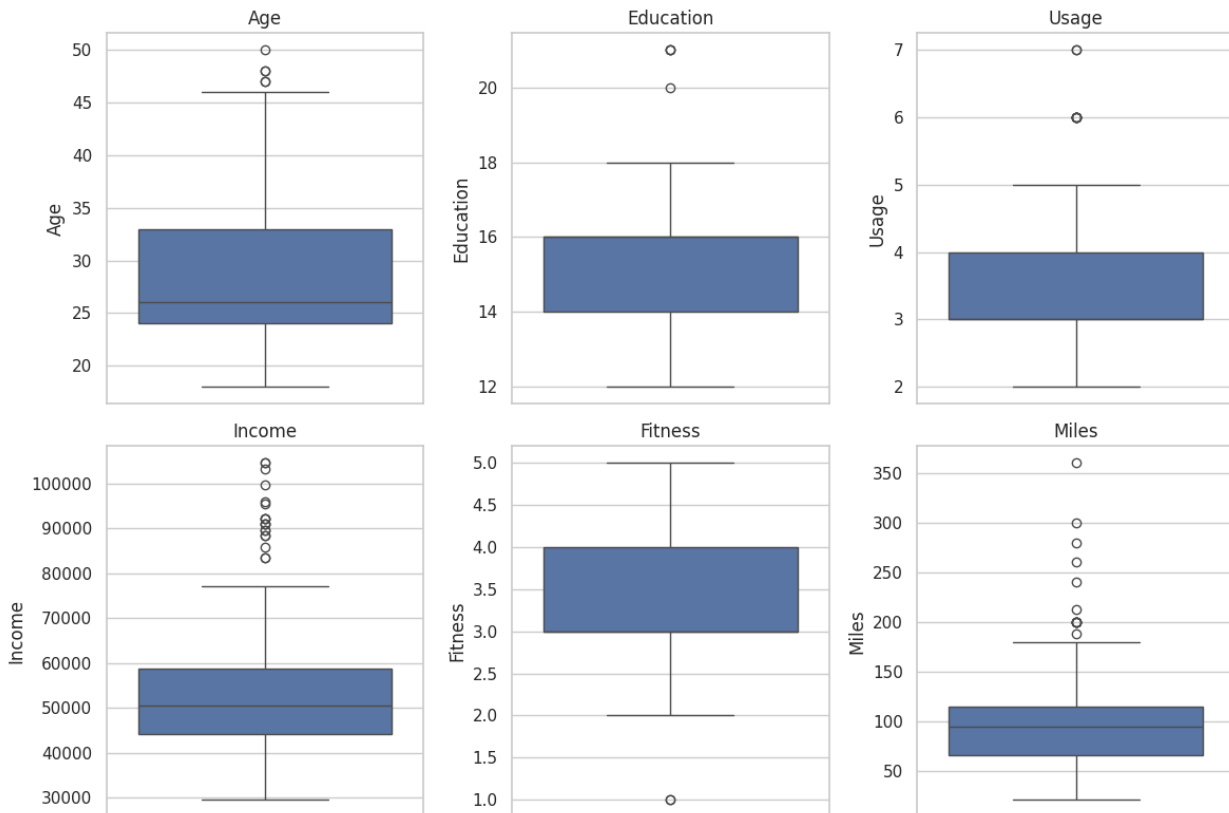
```
(df.isnull().sum()/len(df))*100
```

```
Product          0.0
Age              0.0
Gender           0.0
Education        0.0
MaritalStatus    0.0
Usage            0.0
Fitness          0.0
Income           0.0
Miles            0.0
dtype: float64
```

All variables contain 0% missing values, confirming high data quality.

3.Outlier detection

```
num_cols = ["Age","Education","Usage","Income","Fitness","Miles"]

plt.figure(figsize=(12,8))
for i, col in enumerate(num_cols, 1):
    plt.subplot(2,3,i)
    sns.boxplot(y=df[col])
    plt.title(col)

plt.tight_layout()
plt.show()
```



4.Statistical outlier check

```
df.describe().T[["mean","50%"]]
```

{"summary":"{\n \"name\": \"df\",\n \"rows\": 6,\n \"fields\": [\n {\n \"column\": \"mean\",\n \"properties\": {\n \"dtype\": \"number\",\n \"std\": 21918.35728202715,\n \"min\": 3.311111111111111,\n \"max\": 53719.57777777778,\n \"num_unique_values\": 6,\n \"samples\": [\n 28.788888888888888,\n 15.572222222222223,\n 103.19444444444444\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n },\n {\n \"column\": \"50%\",\n \"properties\": {\n \"dtype\": \"number\",\n \"std\": 20644.368245157482,\n \"min\": 3.0,\n \"max\": 50596.5,\n \"num_unique_values\": 5,\n \"samples\": [\n 16.0,\n 94.0,\n 3.0\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n }\n]\n}","type":"dataframe"}

Outlier Observations:

- Income shows noticeable outliers on the higher side, indicating presence of high-income customers.
- Miles per week also shows high-value outliers representing serious runners.
- Age and Usage show relatively stable distributions with limited extreme values.

- These outliers appear to be genuine customer behavior rather than data errors.

Business Interpretation:

High-income and high-mileage outliers are valuable customer segments rather than anomalies. These customers are likely premium buyers and should be considered an important target audience for advanced treadmill products.

Key Findings:

- Dataset contains no missing values.
- Income and Miles variables show right-skewed distributions.
- Outliers represent high-performance users rather than incorrect data.
- Premium customer segments are visible through extreme values.

5.Business Insights based on Non-Graphical and Visual Analysis

1.Range of Attributes

- Age ranges from young adults to middle-aged customers, showing AeroFit products appeal to a wide age group.
- Income shows a large range, indicating both budget-conscious and high-income customers purchase treadmills.
- Fitness levels range from 1 to 5, meaning products are used by beginners as well as highly fit individuals.
- Weekly usage varies between low and high frequency users, suggesting different exercise commitment levels.
- Miles per week has a wide spread, highlighting the presence of both casual walkers and serious runners.

Overall, AeroFit serves multiple customer segments rather than a single niche market.

2.Distribution of Variables

- Age distribution is concentrated between 20–35 years, indicating strong adoption among young professionals.
- Income distribution is right-skewed, meaning a smaller group of high earners exists alongside many middle-income customers.
- Fitness ratings are centered around level 3, suggesting most buyers consider themselves moderately fit.
- Usage distribution peaks around 3–4 times per week, indicating consistent but realistic exercise habits.
- Miles distribution shows high variability, reflecting differences in workout intensity among customers.

3.Relationship Between Variables

- Fitness and Miles show a strong positive relationship, meaning fitter customers tend to run more weekly miles.
- Usage frequency is positively related to Miles, indicating frequent users perform longer workouts.

- Income indirectly influences product choice, with higher-income customers gravitating toward premium treadmills.
- Age shows limited correlation with product selection, suggesting lifestyle behavior matters more than demographics.

4.Insights from Univariate Analysis

- Histogram of Age shows majority customers are young adults.
- Income histogram reveals right-skewness caused by high-income buyers.
- Fitness distribution indicates moderate fitness dominates the customer base.
- Usage histogram shows customers prefer manageable workout schedules.
- Miles distribution confirms existence of advanced runners alongside beginners.

These plots indicate AeroFit products cater primarily to regular fitness enthusiasts rather than extreme athletes.

5.Insights from Bivariate Analysis

- Income vs Product boxplot shows KP781 buyers have significantly higher income levels.
- Fitness vs Product analysis indicates premium treadmill buyers possess higher fitness ratings.
- Usage vs Product reveals heavy users prefer advanced treadmills.
- Miles vs Product demonstrates that customers expecting higher weekly mileage choose KP781.
- Age vs Product shows minimal variation, suggesting age is not a major factor influencing purchase decisions.

These findings suggest behavioral variables (fitness, usage, mileage) are stronger predictors of product choice than demographic attributes.

6.Overall Business Insight

The analysis reveals three distinct customer segments:

1. Beginner users purchasing entry-level treadmills.
2. Regular fitness users selecting mid-range equipment.
3. High-income and performance-focused users investing in premium treadmills.

Customer behavior and fitness lifestyle play a larger role in product selection than age or demographic characteristics.

6.Business Recommendations

Based on the customer behavior and purchase patterns observed in the analysis, the following actionable recommendations are suggested for AeroFit:

1.Match Treadmills to Fitness: Help customers choose the right treadmill based on how fit they are. Guide beginners to the basic KP281, regular exercisers to the mid-range KP481, and very active people to the advanced KP781. This ensures happy customers and fewer returns.

2.Attract High-End Buyers: People with more money and intense fitness routines often buy the best treadmills. Create special ads for the KP781 that show off its top features and benefits to these wealthier, fitness-focused customers.

3.Market to Young Professionals: Most buyers are younger and exercise regularly. Use online ads and social media to reach young working people who want home fitness equipment.

4.Offer Upgrade Deals: As people use their treadmills more often, they might want a better model. Provide discounts or trade-in options to help KP281 users move up to KP481 or KP781.

5.Encourage Family Fitness: Many buyers are in relationships. Market treadmills as a way for whole families to get fit together, not just for one person.

6.Boost Female Premium Sales: Fewer women are buying the most expensive treadmills. Launch campaigns featuring female athletes and custom workout plans to get more women interested in premium models.

7.Add Extra Perks: Serious treadmill users will appreciate extra services. Offer packages that include workout plans, personal coaching, or fitness app subscriptions to make the treadmills even more appealing and build customer loyalty.

8.Improve Sales Advice: Train sales staff to ask simple questions like: 'How often will you exercise?' 'What are your fitness goals?' 'How many miles do you plan to run?' Their answers will help staff recommend the perfect treadmill.

Conclusion

The analysis demonstrates that AeroFit serves multiple customer segments with different fitness needs and purchasing power. Entry-level treadmills attract mass customers, while premium models appeal to high-income and performance-focused users. Understanding customer behavior enables AeroFit to improve product recommendations, marketing efficiency, and customer satisfaction.